HADROS3: Physics and Statistical Theory of the Implemented Pipeline

Rafael C. R. de Lima

Implemented theory through H3-W8b Observer Image Branch Analysis and H3-W9b POWHEG real-smoke/free execution

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This document describes only the physics, statistics, numerical methods, and proxy models implemented in the current HADROS3 pipeline. It is not a user manual, dashboard guide, installation guide, or proposal for future stages. Where the implementation uses diagnostic renderers or physics proxies, that status is stated explicitly. This is a living theory document for HADROS3. It describes exclusively the physics currently implemented in the official HADROS3 pipeline. Future or planned models are not described here until they become part of the official implementation.

Official Physics Reference

This document is the official reference describing the physics currently implemented in HADROS3.

If any discrepancy is found between the implementation and this document, either:

- the implementation must be corrected; or
- this document must be updated.

Such discrepancies must never be ignored.

The scientific traceability chain for implemented physics is

$$\text{code} \rightarrow \text{provenance} \rightarrow \text{Theory PDF}. \tag{0.1}$$

Each simulation provenance record must identify the theory version that describes its physical and statistical assumptions.

Theory Version Information

Field	Value
Software Version	0.9.0
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Theory Version	1.2
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Stage	Theory status
H3-W5 UHE Source	implemented and documented
H3-W6 Forward Geodesics	implemented and documented
H3-W7 DIS Interaction Sampler	implemented and documented
H3-W8 Observer Bridge	implemented/proxy and documented
H3-W8b Observer Image Branch Analysis	implemented/proxy and documented
H3-W9a POWHEG Dry Run	implemented/dry run and documented
H3-W9b POWHEG Real Smoke/Free	implemented/smoke+free and documented
H3-W10 PYTHIA	planned, not documented as implemented
H3-W11 GEANT4	planned, not documented as implemented
H3-W12 Photon Transport	planned, not documented as implemented
H3-W13 Spectra	planned, not documented as implemented

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Scope and Notation

The implemented HADROS3 pipeline is organized as a staged calculation,

$$\text{inputs} \to \text{run()} \to \text{outputs} \to \text{diagnostics} \to \text{provenance}. \tag{0.2}$$

Each implemented stage consumes official products from earlier stages and writes to its own output directory. The stages covered here are the camera preview, Kerr geometry utilities, analytic medium, UHE source, forward null geodesics, DIS optical-depth sampler, Observer Bridge scoring, and POWHEG dry-run hard-process preparation.

This document is maintained as a required living reference for implemented physics. When a new HADROS3 stage changes or adds physical equations, statistical weights, physical approximations, or proxy models, this source file and the regenerated PDF must be updated with the implementation.

Table 3: Implemented-stage coverage in this theory document. Planned stages are listed only to mark that their physics is not yet documented as implemented.

HADROS3 stage	Implementation status	Theory documented
H3-W5 UHE Source	implemented	yes
H3-W6 Forward Geodesics	implemented	yes
H3-W7 DIS Interaction Sampler	implemented	yes
H3-W8 Observer Bridge	implemented/proxy	yes
H3-W8b Observer Image Branch Analysis	implemented/proxy	yes
H3-W9a POWHEG Dry Run	implemented/dry run	yes
H3-W9b POWHEG Real Smoke/Free	implemented/smoke+free	yes
H3-W10 PYTHIA	planned	no
H3-W11 GEANT4	planned	no
H3-W12 Photon Transport	planned	no
H3-W13 Spectra	planned	no

All radii in the dynamical Kerr calculation are expressed in gravitational radius units,

$$r_g = \frac{GM}{c^2}. \tag{0.3}$$

The implementation uses Boyer-Lindquist coordinates $x^\mu = (t, r, \theta, \phi)$ and the metric signature $(-, +, +, +)$. The dimensionless Kerr spin is denoted by a in units of r_g. The code clamps spin-dependent operations away from the extremal singular limit where needed.

Table 4: Primary symbols used in the implemented theory.

Symbol	Meaning
r, θ, ϕ	Boyer-Lindquist spherical coordinates in r_g units.
Σ, Δ, A	Standard Kerr metric functions.
p_μ	Covariant four-momentum components.
$E_{\nu,\mathrm{local}}$	Neutrino energy measured in a local medium or ZAMO frame.
$\rho(r, \theta)$	Analytic torus rest-mass density.
n_b	Baryon number density, $n_b = \rho/m_b$.
$\sigma_{\nu N}(E)$	Tabulated charged-current neutrino-nucleon DIS cross section.
τ	Integrated DIS optical depth along one forward path.
S_{obs}	Observer Bridge final observation score.
x	Bjorken momentum fraction of the struck parton.
y	DIS inelasticity, $y = P \cdot q/(P \cdot k)$.
Q^2	Virtuality of the exchanged boson, $Q^2 = -q^2 > 0$.
W	Invariant mass of the produced hadronic system.
$\sqrt{s}$	Center-of-mass energy of the neutrino–nucleon system.
G_F	Fermi constant.
m_W	W boson mass.
$V_{q_i q_j}$	CKM matrix element for the quark flavor transition $q_i \to q_j$.
F_2, xF_3, F_L	CC neutrino DIS structure functions.
$Y_\pm$	DIS angular factors, $Y_\pm = 1 \pm (1-y)^2$.
$B(\Phi_n)$	Born differential cross section on n-body phase space.
$\bar{B}(\Phi_n)$	NLO-corrected Born (virtual + integrated real counterterms).
$\Delta(\Phi_n, p_T)$	POWHEG Sudakov form factor.

Chapter 1

Camera / Kerr Ray Tracing

1.1 Implemented camera modes

The camera preview layer is a visual and diagnostic stage. It is separate from the DIS interaction physics and from the Observer Bridge scoring. Its main product is a rendered observer-view image, typically `CameraPreview/hadros3_camera_preview.png`. The implemented backend selection includes:

- `analytic_geometry_only`: always available; a schematic camera preview, not a physical Kerr ray-traced image.
- `kerr_like_cuda`: a self-contained HADROS3 CUDA preview executable using a Kerr-like camera model.
- `full_kerr`: a self-contained HADROS3 CUDA preview executable using the full-Kerr preview mode when available.
- `legacy_cpu_geodesic_preview`: detected for provenance but not used as the primary self-contained HADROS3 runtime path.

The camera preview can render a torus-like visual object and polar funnels. This visual object is not the DIS optical-depth calculation. The provenance marks camera-preview products separately from the shared `MediumRenderer`. In current provenance, the camera preview records `medium_renderer_used = false` because its purpose is visual camera preview, not the shared diagnostic density-map renderer.

1.2 Camera frame

Observer Bridge camera diagnostics use the same geometric convention for the camera direction that is also relevant when overlaying candidates on the camera preview. A camera position is built from

$$\mathbf{x}_{\mathrm{obs}} = r_{\mathrm{obs}} \begin{pmatrix} \sin i \cos \varphi_{\mathrm{obs}} \\ \sin i \sin \varphi_{\mathrm{obs}} \\ \cos i \end{pmatrix}, \tag{1.1}$$

where r_{obs} is `observer_distance_rg`, i is `observer_inclination_deg`, and φ_{obs} is `observer_azimuth_deg`. The forward axis points approximately toward the origin,

$$\hat{\mathbf{f}} = -\frac{\mathbf{x}_{\mathrm{obs}}}{|\mathbf{x}_{\mathrm{obs}}|}. \tag{1.2}$$

The right and up axes are built from a world z axis, with a fallback right axis if the camera is too close to the polar direction:

$$\hat{\mathbf{r}}_{\text{cam}} = \frac{\hat{\mathbf{f}} \times \hat{\mathbf{z}}}{|\hat{\mathbf{f}} \times \hat{\mathbf{z}}|}, \tag{1.3}$$

$$\hat{\mathbf{u}}_{\text{cam}} = \frac{\hat{\mathbf{r}}_{\text{cam}} \times \hat{\mathbf{f}}}{|\hat{\mathbf{r}}_{\text{cam}} \times \hat{\mathbf{f}}|}. \tag{1.4}$$

1.3 Pinhole diagnostic projection

For Observer Bridge candidate overlays, the implemented projection is a geometric pinhole proxy. It converts a candidate position $\mathbf{x}$ into a camera vector

$$\mathbf{d} = \mathbf{x} - \mathbf{x}_{\text{obs}} \tag{1.5}$$

and projects it as

$$z_{\text{cam}} = \mathbf{d} \cdot \hat{\mathbf{f}}, \tag{1.6}$$

$$x_{\text{ndc}} = \frac{\mathbf{d} \cdot \hat{\mathbf{r}}_{\text{cam}}}{z_{\text{cam}} \tan(\text{FOV}_x/2)}, \tag{1.7}$$

$$y_{\text{ndc}} = \frac{\mathbf{d} \cdot \hat{\mathbf{u}}_{\text{cam}}}{z_{\text{cam}} \tan(\text{FOV}_y/2)}. \tag{1.8}$$

The field-of-view flag is true when $z_{\text{cam}} > 0$ and $|x_{\text{ndc}}| \leq 1$, $|y_{\text{ndc}}| \leq 1$. The camera overlay records `candidate_overlay_projection_model = geometric_pinhole_proxy` and `candidate_overlay_not_r` `= true`. Therefore the background image may be generated by the camera renderer, but the plotted Observer Bridge candidates are not ray-traced secondary particles.

1.4 Inputs and outputs

Role	Implemented item
Inputs	Black-hole spin, observer distance, inclination, azimuth, FOV, image resolution, torus and funnel visual parameters.
Outputs	`CameraPreview/hadros3_camera_preview.png`, `CameraPreview/hadros3_camera_preview_summary.json`.
Approximations	Analytic fallback is schematic. CUDA preview is visual camera rendering; candidate overlays remain pinhole proxies.
Provenance	Records backend used, fallback status, whether CUDA was used, and whether full-Kerr preview mode was requested.

Chapter 2

Black Hole and Kerr Geometry

2.1 Metric functions

HADROS3 uses the Kerr metric in Boyer-Lindquist coordinates and in r_g units. The standard metric functions are

$$\Sigma = r^2 + a^2 \cos^2\theta, \tag{2.1}$$
$$\Delta = r^2 - 2r + a^2, \tag{2.2}$$
$$A = (r^2 + a^2)^2 - a^2 \Delta \sin^2\theta. \tag{2.3}$$

The outer horizon radius is

$$r_+ = 1 + \sqrt{1 - a^2}. \tag{2.4}$$

The nonzero covariant metric components implemented for the dynamical calculation are

$$g_{tt} = -\left(1 - \frac{2r}{\Sigma}\right), \tag{2.5}$$
$$g_{t\phi} = -\frac{2ar\sin^2\theta}{\Sigma}, \tag{2.6}$$
$$g_{rr} = \frac{\Sigma}{\Delta}, \tag{2.7}$$
$$g_{\theta\theta} = \Sigma, \tag{2.8}$$
$$g_{\phi\phi} = \left(r^2 + a^2 + \frac{2a^2 r \sin^2\theta}{\Sigma}\right)\sin^2\theta = \frac{A\sin^2\theta}{\Sigma}. \tag{2.9}$$

The corresponding inverse components used by the Hamiltonian geodesic solver are

$$g^{tt} = -\frac{A}{\Sigma\Delta}, \tag{2.10}$$
$$g^{t\phi} = -\frac{2ar}{\Sigma\Delta}, \tag{2.11}$$
$$g^{rr} = \frac{\Delta}{\Sigma}, \tag{2.12}$$
$$g^{\theta\theta} = \frac{1}{\Sigma}, \tag{2.13}$$
$$g^{\phi\phi} = \frac{\Delta - a^2\sin^2\theta}{\Sigma\Delta\sin^2\theta}. \tag{2.14}$$

2.2 ZAMO quantities

The implemented ZAMO lapse and frame-dragging angular velocity are

$$\alpha = \sqrt{\frac{\Sigma\Delta}{A}}, \tag{2.15}$$

$$\omega = \frac{2ar}{A}. \tag{2.16}$$

For a covariant momentum with components p_t and p_ϕ, the ZAMO energy proxy used in the C++ tetrad utilities is

$$E_{\mathrm{ZAMO}} = -\frac{p_t + \omega p_\phi}{\alpha}. \tag{2.17}$$

The local tetrad initialization maps a normalized local spatial direction (n_r, n_θ, n_ϕ) into a contravariant null vector and then lowers the index with $g_{\mu\nu}$. The implementation checks that the resulting null norm is finite and that the ZAMO energy is positive.

2.3 Null invariant

The null condition is

$$g^{\mu\nu} p_\mu p_\nu = 0. \tag{2.18}$$

Both Python reference utilities and the C++ forward-geodesic backend monitor this invariant. The calculation also monitors the Killing quantities

$$E_\infty = -p_t, \tag{2.19}$$

$$L_z = p_\phi, \tag{2.20}$$

which are conserved because the metric is stationary and axisymmetric.

Chapter 3

Torus / Medium

3.1 Analytic density model

The implemented medium model is an analytic torus density field. It is not a hydrodynamical simulation. The density is evaluated as

$$\rho(r,\theta) = \rho_0 \exp\left[-\frac{1}{2}\left(\frac{r - r_{\mathrm{peak}}}{\sigma_r}\right)^2\right] \exp\left[-\frac{1}{2}\left(\frac{\theta - \pi/2}{\sigma_\theta}\right)^2\right], \tag{3.1}$$

with the hard radial support

$$\rho(r,\theta) = 0 \quad \text{if} \quad r < r_{\mathrm{inner}} \quad \text{or} \quad r > r_{\mathrm{outer}}. \tag{3.2}$$

The implemented radial width is

$$\sigma_r = \frac{1}{2}(r_{\mathrm{outer}} - r_{\mathrm{inner}}), \tag{3.3}$$

and the angular width is

$$\sigma_\theta = \max(\theta_{\mathrm{half}}, 10^{-6}). \tag{3.4}$$

The configuration parameter `half_opening_angle_deg` is therefore a Gaussian angular width, not a hard angular boundary. This interpretation is recorded in diagnostics as `half_opening_angle_interpreta` `= gaussian_width_not_boundary`.

Within the hard radial support, the code can apply a configured `density_floor_g_cm3` by taking the maximum of the analytic value and the floor. Outside the radial shell, the density remains exactly zero.

3.2 MediumRenderer

The shared `MediumRenderer` renders the same analytic density model used by the DIS sampler diagnostics. It records

$$\texttt{medium\_renderer\_used} = \texttt{true}, \tag{3.5}$$

$$\texttt{density\_model\_has\_hard\_radial\_cut} = \texttt{true}, \tag{3.6}$$

$$\texttt{density\_model\_theta\_profile} = \texttt{gaussian}, \tag{3.7}$$

$$\texttt{density\_model\_theta\_is\_hard\_cut} = \texttt{false}. \tag{3.8}$$

The renderer draws the hard radial cuts and Gaussian angular density levels. Those angular levels are visual density contours, not physical walls.

3.3 Inputs and outputs

Role	Implemented item
Inputs	`rho0_g_cm3`, `r_inner_rg`, `r_peak_rg`, `r_outer_rg`, `half_opening_angle_deg`, optional density floor.
Used by	DIS density evaluation, DIS density diagnostics, forward geometry diagnostics, Observer Bridge diagnostic projections.
Approximation	Static analytic density field; no GRMHD evolution; no self-consistent thermal or composition model.
Output semantics	Positive density only inside the hard radial shell; Gaussian tail in θ.

Chapter 4

UHE Source

4.1 Coordinate cone sampling

The implemented UHE source is a coordinate-volume sampler in a polar cone. For a cone spanning $r_{\rm min} \le r \le r_{\rm max}$ and $\theta_{\rm min} \le \theta \le \theta_{\rm max}$, the coordinate volume used for sampling is

$$V_{\rm cone} = \frac{2\pi}{3}\left(r_{\rm max}^3 - r_{\rm min}^3\right)\left(\cos\theta_{\rm min} - \cos\theta_{\rm max}\right). \tag{4.1}$$

Uniform coordinate-volume sampling is implemented by drawing $u_r, u_\theta, u_\phi \sim U(0,1)$ and setting

$$r = \left[r_{\rm min}^3 + u_r(r_{\rm max}^3 - r_{\rm min}^3)\right]^{1/3}, \tag{4.2}$$
$$\cos\theta = \cos\theta_{\rm min} - u_\theta(\cos\theta_{\rm min} - \cos\theta_{\rm max}), \tag{4.3}$$
$$\phi = 2\pi u_\phi. \tag{4.4}$$

The source sampling and physical PDFs are both recorded as

$$p_{\rm source} = \frac{1}{V_{\rm cone}}, \tag{4.5}$$

so the implemented source weight is unity for the default model:

$$w_{\rm source} = \frac{p_{\rm physical}}{p_{\rm sampling}} = 1. \tag{4.6}$$

4.2 Energy and directions

The implemented source energy model is monoenergetic:

$$E_{\nu,\rm emit} = E_0. \tag{4.7}$$

For the isotropic local direction model, the code samples a unit vector in a ZAMO orthonormal frame:

$$\cos\alpha = 2u_\alpha - 1, \tag{4.8}$$
$$\beta = 2\pi u_\beta, \tag{4.9}$$
$$n_r = \cos\alpha, \tag{4.10}$$
$$n_\theta = \sin\alpha\cos\beta, \tag{4.11}$$
$$n_\phi = \sin\alpha\sin\beta. \tag{4.12}$$

The sampling PDF is

$$p_\Omega = \frac{1}{4\pi}, \tag{4.13}$$

and the default direction weight is unity.

The source stage does not itself produce the physical Kerr four-momentum for forward propagation. It records the source position, energy, and local direction. The forward-geodesic stage constructs the physical Kerr null momentum from those quantities.

4.3 Inputs and outputs

Role	Implemented item
Inputs	Cone geometry, sample count, random seed, monoenergetic neutrino energy, direction model.
Outputs	`UHEsource/uhe_neutrino_source_samples.jsonl` and source summary products.
Weights	`source_weight = 1` and `direction_weight = 1` for the implemented default uniform/properly matched sampling.
Approximation	Coordinate cone volume, not a GR invariant emissivity integral.

Chapter 5

Forward Geodesics

5.1 Momentum construction

The forward stage consumes UHE source samples and constructs a Kerr null covector. For a local isotropic direction, the ZAMO tetrad maps the local direction into a null four-vector and then into covariant momentum components. For legacy coordinate radial directions, the implementation constructs a covariant momentum with $p_t = -E$ and solves the null constraint for p_r.

The implementation records that the physical Kerr four-momentum is constructed in the forward stage, not in the source sampler.

5.2 Hamiltonian system

The Hamiltonian for null geodesics is

$$H(x, p) = \frac{1}{2} g^{\mu\nu}(x) p_\mu p_\nu. \tag{5.1}$$

The implemented equations of motion are

$$\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} = g^{\mu\nu} p_\nu, \tag{5.2}$$

$$\frac{\mathrm{d}p_i}{\mathrm{d}\lambda} = -\frac{1}{2} \partial_i g^{\mu\nu} p_\mu p_\nu, \qquad i \in \{r, \theta\}. \tag{5.3}$$

The cyclic momenta are conserved by construction in the differential system:

$$\frac{\mathrm{d}p_t}{\mathrm{d}\lambda} = 0, \qquad \frac{\mathrm{d}p_\phi}{\mathrm{d}\lambda} = 0. \tag{5.4}$$

The current implementation uses an RK4 stepping procedure with substeps and invariant checks. Derivatives of inverse metric components are evaluated numerically. When needed, p_r can be renormalized to keep the null constraint within tolerance. The solver stops when it reaches the horizon neighborhood, the configured outer radius, maximum steps, or an invalid invariant condition.

5.3 Segment quantities

Each accepted forward path is written as a set of path segments. The segment length used downstream is a coordinate path-length proxy,

$$\Delta l_{\mathrm{coord}} = \left[(\Delta r)^2 + (r_{\mathrm{mid}}\Delta\theta)^2 + (r_{\mathrm{mid}} \sin\theta_{\mathrm{mid}}\Delta\phi)^2\right]^{1/2}. \tag{5.5}$$

This is then converted to cm in the DIS stage by multiplying by r_g in cm. This is an implemented optical-depth path-length approximation and should not be confused with a fully covariant matter-frame line element.

5.4 Inputs and outputs

Role	Implemented item
Inputs	`UHEsource/uhe_neutrino_source_samples.jsonl`, black-hole spin, step settings, tolerances, outer radius, requested number of samples.
Outputs	`ForwardGeodesics/uhe_neutrino_forward_paths.jsonl`, `ForwardGeodesics/uhe_neutrino_forward_path_segments.jsonl`, and summary diagnostics.
Invariants	Null norm, Killing energy, and L_z are monitored.
Approximation	Numerical RK4 Hamiltonian evolution with coordinate segment lengths for downstream optical depth.

Chapter 6

DIS Interaction Sampler

6.1 Density and local energy

The DIS sampler consumes forward path segments and evaluates the analytic torus density at segment midpoints. The baryon density is

$$n_b(r, \theta) = \frac{\rho(r, \theta)}{m_b}, \tag{6.1}$$

where m_b is the baryon mass in grams.

The local neutrino energy is computed from the covariant momentum. If the configured static medium model is valid at that position, the code uses the static observer expression

$$E_{\text{local}} = -p_t u^t, \qquad u^t = \frac{1}{\sqrt{-g_{tt}}}. \tag{6.2}$$

If a static observer is not valid, or if the implementation needs the fallback, the ZAMO expression is used:

$$E_{\text{local}} = -\left(\frac{p_t}{\alpha} + \frac{\omega p_\phi}{\alpha}\right) = -\frac{p_t + \omega p_\phi}{\alpha}. \tag{6.3}$$

The static-to-ZAMO fallback is recorded as a physics-risk approximation.

6.2 Cross-section interpolation

The DIS charged-current cross sections are read from compact tabulated files for the implemented GBW and IIM models. If an energy lies inside the tabulated range, the interpolation is log-log:

$$\ln \sigma_{\nu N}(E) = \ln \sigma_0 + \frac{\ln E - \ln E_0}{\ln E_1 - \ln E_0} (\ln \sigma_1 - \ln \sigma_0). \tag{6.4}$$

Requests outside the tabulated range are rejected for that segment and the segment contribution is set to zero in the diagnostic recomputation.

6.3 Optical depth and probability

For each segment,

$$\Delta\tau_i = n_{b,i}\, \sigma_{\nu N}(E_{\text{local},i})\, \Delta l_i, \tag{6.5}$$

where Δl_i is the coordinate path-length proxy converted from r_g to cm. The total optical depth along a path is

$$\tau = \sum_i \Delta\tau_i. \tag{6.6}$$

The interaction probability is

$$P_{\rm int} = 1 - \exp(-\tau), \tag{6.7}$$

implemented numerically as

$$P_{\rm int} = -\,\mathrm{expm1}(-\tau) \tag{6.8}$$

with clipping to the interval $[0, 1]$.

6.4 Interaction sampling

If a path is accepted by the Bernoulli trial with probability $P_{\rm int}$, the segment index is sampled by the inverse CDF

$$P(i) = \frac{\Delta\tau_i}{\sum_j \Delta\tau_j}. \tag{6.9}$$

After the segment is selected, the implemented accepted-point sampler requires the final recorded interaction point to have positive medium density. It performs rejection sampling inside the chosen segment and records

$$\texttt{interaction\_point\_density\_checked} = \texttt{true}, \tag{6.10}$$

$$\texttt{interaction\_point\_inside\_medium} = \texttt{true} \quad \text{when} \quad \rho(r_{\rm int}, \theta_{\rm int}) > 0, \tag{6.11}$$

$$\texttt{interaction\_point\_rho\_g\_cm3} = \rho(r_{\rm int}, \theta_{\rm int}). \tag{6.12}$$

The fallback policy is recorded in `interaction_point_sampling_method`. The present criterion for a valid accepted point is $\rho(r_{\rm int}, \theta_{\rm int}) > 0$.

6.5 Weights

The DIS products preserve source and direction weights from the UHE source. The pre-event weight used downstream is recorded as `final_pre_event_weight`. The DIS sampler also records optical-depth and expected-interaction quantities so that later stages can distinguish the sampling decision from pre-event observation scoring.

6.6 Inputs and outputs

Role	Implemented item
Inputs	Forward geodesic paths and segments, source samples, DIS cross-section tables, analytic medium parameters, random seed.
Outputs	`DIS/dis_path_optical_depths.jsonl`, `DIS/dis_accepted_interactions.jsonl`, optical-depth report, diagnostics, GBW/IIM comparison products.

Diagnostics	Tau distribution, interaction probability distribution, optical depth maps, medium density map, accepted interaction distributions, sigma and correlation plots.
Approximation	Static or ZAMO local energy model; coordinate path-length proxy; compact cross-section tables.

Chapter 7

Observer Bridge

7.1 Purpose

The Observer Bridge consumes accepted DIS interactions and assigns a non-destructive observation score. It does not generate secondary particles, does not call POWHEG, does not call PYTHIA, does not call GEANT4, and does not perform photon transport. All accepted DIS interactions become Observer Bridge candidates, even if they are outside the camera FOV or have small score.

7.2 Geometry proxies

The interaction position is converted to Cartesian coordinates through

$$(x, y, z) = (r \sin\theta \cos\phi,\, r \sin\theta \sin\phi,\, r \cos\theta)\,. \tag{7.1}$$

The observer is placed at the configured camera position. The geometric direction from the interaction to the observer is

$$\hat{\mathbf{n}}_{\text{obs}} = \frac{\mathbf{x}_{\text{obs}} - \mathbf{x}_{\text{int}}}{|\mathbf{x}_{\text{obs}} - \mathbf{x}_{\text{int}}|}. \tag{7.2}$$

The implemented escape proxy compares this direction with the local radial direction,

$$w_{\text{esc}} = \max\left(0,\, \hat{\mathbf{x}}_{\text{int}} \cdot \hat{\mathbf{n}}_{\text{obs}}\right). \tag{7.3}$$

This is a proxy, not a particle-transport escape calculation.

The FOV weight is either a hard FOV flag or a soft Gaussian-like angular proxy, depending on configuration. The visibility, line-of-sight, and redshift models are explicitly recorded as proxies. The implemented default redshift and line-of-sight weights are unity proxies unless configured otherwise.

7.3 Score definition

The implemented physics weight in the C++ Observer Bridge is

$$w_{\text{physics}} = \texttt{final\_pre\_event\_weight}, \tag{7.4}$$

with a fallback to source times direction weight when that field is absent. The observer weight is the product of the implemented proxy factors:

$$w_{\text{observer}} = w_{\text{esc}}\, w_{\text{visibility}}\, w_{\text{FOV}}\, w_{\text{distance}}\, w_{\text{redshift}}\, w_{\text{LOS}}. \tag{7.5}$$

The final observation score is

$$S_{\mathrm{obs}} = w_{\mathrm{physics}}\, w_{\mathrm{observer}}. \tag{7.6}$$

All three quantities are recorded for each candidate: `physics_weight`, `observer_weight`, and `final_observation_score`. The implementation enforces non-negative proxy weights by clipping or construction.

7.4 Proxy risk flags

The reports record the proxy nature of the bridge:

$$\texttt{secondary\_particle\_proxy\_model} \neq \texttt{full\_transport}, \tag{7.7}$$

$$\texttt{escape\_proxy\_physics\_risk} = \texttt{true}, \tag{7.8}$$

$$\texttt{visibility\_proxy\_physics\_risk} = \texttt{true}, \tag{7.9}$$

$$\texttt{redshift\_proxy\_physics\_risk} = \texttt{true}. \tag{7.10}$$

The stage status is scoring-only. It is intentionally non-destructive.

7.5 Inputs and outputs

Role	Implemented item
Inputs	`DIS/dis_accepted_interactions.jsonl` and runtime configuration. It may consult previous products for diagnostics.
Outputs	`ObserverBridge/observer_bridge_candidates.jsonl`, `ObserverBridge/observer_bridge_ranked_events.jsonl`, JSON/CSV summaries, camera-view and overlay diagnostics.
Invariant policy	Number of candidates scored equals number of accepted DIS interactions consumed. FOV is not used as a destructive filter.
Approximation	Geometric escape, visibility, FOV, redshift, and line-of-sight proxies. No secondary-particle generation.

Chapter 8

Observer Image Branch Analysis

H3-W8b introduces an explicit observational image-branch analysis between the Observer Bridge and POWHEG. The Observer Bridge remains a non-destructive candidate scorer: it maps every accepted DIS interaction into an observational candidate and records a closest Kerr-ray diagnostic. H3-W8b asks a different question. Instead of assigning

$$\text{candidate} \to \arg\min_{\text{ray}} d_{\text{closest}}, \tag{8.1}$$

it groups all compatible camera rays into observed image branches and selects a primary branch by an auditable proxy score.

8.1 Inputs

The implemented stage consumes official Observer Bridge products:

$$\texttt{ObserverBridge/observer\_candidate\_kerr\_pixel\_map.jsonl}, \tag{8.2}$$
$$\texttt{ObserverBridge/observer\_bridge\_selected\_candidates.jsonl}, \tag{8.3}$$
$$\texttt{ObserverBridge/observer\_bridge\_ranked\_events.jsonl}. \tag{8.4}$$

When available, it also consumes the Observer Bridge multi-image audit product `ObserverBridge/candidate_mu` which records sampled camera rays passing within the configured matching radius of each candidate. No DIS interaction, forward geodesic, camera preview, or Kerr integrator result is modified by H3-W8b.

8.2 Image branch definition

For a candidate c, let $\mathcal{R}_c$ be the set of sampled camera rays whose minimum spatial separation from the candidate is below the matching tolerance. Rays are clustered in image-pixel space. Each connected cluster is an image branch $B_{c,j}$ and records:

$$B_{c,j} = \left\{ N_{\text{rays}}, \bar{x}_{\text{pix}}, \bar{y}_{\text{pix}}, \text{Cov}(x_{\text{pix}}, y_{\text{pix}}), \bar{d}_{\text{closest}}, d_{\min}, d_{\max}, s_{\text{pix}} \right\}. \tag{8.5}$$

Here N_{rays} is the number of compatible rays, $(\bar{x}_{\text{pix}}, \bar{y}_{\text{pix}})$ is the branch centroid on the camera preview pixel plane, and s_{pix} is the RMS pixel spread around the centroid.

8.3 Branch score proxy

The primary observed branch is not selected by the single closest ray. Instead, H3-W8b assigns each branch a proxy score

$$S_{\text{branch}} = w_{\text{rays}}\, w_{\text{closest}}\, w_{\text{compactness}}, \tag{8.6}$$

with the implemented proxy weights

$$w_{\text{rays}} = N_{\text{rays}}, \tag{8.7}$$

$$w_{\text{closest}} = \frac{1}{\max(\bar{d}_{\text{closest}}, \epsilon)}, \tag{8.8}$$

$$w_{\text{compactness}} = \frac{1}{\max(s_{\text{pix}}, 1\,\text{pixel})}. \tag{8.9}$$

This score favors image branches supported by many rays, small mean closest approach, and compact clusters on the image plane. It is a ranking proxy, not a full magnification, radiative transfer, or detector-response calculation.

8.4 Primary branch selection

For each candidate, H3-W8b selects

$$B_{c,\text{primary}} = \arg\max_j S_{\text{branch}}(B_{c,j}). \tag{8.10}$$

The selected primary branches are written to `ObserverImageBranches/observer_image_primary_branches.js` Downstream POWHEG preparation consumes that file, so POWHEG receives one primary observed image branch per selected Observer Bridge candidate and does not perform its own image-branch selection.

8.5 Outputs and provenance

The stage writes exclusively under `ObserverImageBranches/`: `observer_image_branches.jsonl`, `observer_image_primary_branches.jsonl`, `observer_image_branch_summary.json`, `observer_image_branc` `observer_image_statistics.json`, diagnostic PNGs, a CSV table, and `observer_branch_view.html`. Provenance records `observer_image_branch_analysis_invoked=true`, `branch_scoring_model=ray_count_cl` `primary_branch_selection_model=argmax_branch_score`, `mean_branches_per_candidate`, `fraction_multi` and `primary_branch_selection_proxy=true`.

8.6 Physical limitations

The current branch score is observationally motivated but approximate. It does not compute rigorous Kerr caustic magnification, phase-space beam Jacobians, secondary-particle emission, absorption, or detector response. It should be treated as a transparent primary-image proxy until a later stage implements a more complete lensing and transport model.

Chapter 9

POWHEG Hard Process Preparation and Smoke Execution

9.1 Hard subprocess: charged-current neutrino DIS

The implemented POWHEG process is `nudis` from POWHEG-BOX-RES, which simulates charged-current (CC) deep-inelastic scattering of a neutrino on a nucleon at next-to-leading order (NLO) in QCD. The Born-level hard subprocess is

$$\nu_\ell(k) + q_i(p) \;\longrightarrow\; \ell^-(k') + q_j(p'), \tag{9.1}$$

mediated by a virtual W^+ boson with four-momentum transfer $q^\mu = k^\mu - k'^\mu$. The incoming neutrino carries the full UHE energy $E_{\nu,\mathrm{local}}$ from the accepted DIS interaction point; the target is a parton q_i drawn from the nucleon PDF with momentum fraction x. HADROS3 specifies the two beam particles as

$$\texttt{ih1} = 12 \quad (\nu_e,\ \text{PDG electron neutrino}), \tag{9.2}$$

$$\texttt{ih2} = 1 \quad (\text{proton/nucleon, PDF target}). \tag{9.3}$$

The process is selected in the POWHEG input card by `channel_type`$= 3$ (CC channel) and `vtype`$= 2$ (specific CC flavor configuration in the `nudis` implementation).

9.2 DIS kinematic variables

The standard DIS kinematic invariants are built from the neutrino four-momentum k^μ, the nucleon four-momentum P^μ, and the momentum-transfer four-momentum $q^\mu = k^\mu - k'^\mu$:

$$s = (k + P)^2 \approx 2\, m_N\, E_{\nu,\mathrm{local}}, \tag{9.4}$$

$$Q^2 = -q^2 > 0, \tag{9.5}$$

$$x = \frac{Q^2}{2\, P \cdot q}, \tag{9.6}$$

$$y = \frac{P \cdot q}{P \cdot k} = \frac{Q^2}{xs}, \tag{9.7}$$

$$W^2 = (P + q)^2 = m_N^2 + Q^2 \frac{1 - x}{x}. \tag{9.8}$$

The approximation for s in equation (9.4) holds at UHE energies where $E_{\nu,\mathrm{local}} \gg m_N$. The Bjorken variable $x \in (0, 1]$ is the momentum fraction of the struck parton, $y \in (0, 1]$ is the inelasticity (fractional energy transfer to the hadronic system in the nucleon rest frame), and W is the invariant mass of the produced hadronic system.

9.3 CC neutrino–nucleon DIS cross section

The double-differential cross section for inclusive $\nu_\ell N$ CC DIS is expressed through three structure functions F_2, F_L, and xF_3:

$$\frac{\mathrm{d}^2\sigma^{\mathrm{CC}}}{\mathrm{d}x\,\mathrm{d}y} = \frac{G_F^2\,s}{4\pi}\left(\frac{m_W^2}{m_W^2+Q^2}\right)^2 \left[Y_+\,F_2(x,Q^2) - y^2 F_L(x,Q^2) + Y_-\,xF_3(x,Q^2)\right], \tag{9.9}$$

where G_F is the Fermi constant, $m_W \simeq 80.4\,\mathrm{GeV}$ is the W boson mass, and the inelasticity factors are

$$Y_\pm = 1 \pm (1-y)^2. \tag{9.10}$$

The sign in front of xF_3 is $+Y_-$ for neutrinos (as written) and $-Y_-$ for antineutrinos. The propagator factor $(m_W^2/(m_W^2+Q^2))^2$ becomes significant when $Q^2 \gtrsim m_W^2$.

At leading order in QCD the structure functions are expressed through the parton distributions weighted by CKM elements. For CC νN DIS the struck parton must be a down-type quark $q_i \in \{d,s,b\}$ (or an up-type antiquark from the sea), and the relevant combinations are

$$F_2^{\mathrm{CC}}(x,Q^2) = 2x\sum_i \left|V_{q_i q_j}\right|^2 f_{q_i}(x,Q^2), \tag{9.11}$$

$$xF_3^{\mathrm{CC}}(x,Q^2) = 2x\sum_i \left|V_{q_i q_j}\right|^2 \mathrm{sign}(q_i)\, f_{q_i}(x,Q^2), \tag{9.12}$$

where $V_{q_i q_j}$ is the CKM element for the transition $q_i \to q_j$ and $\mathrm{sign}(q_i) = +1$ (-1) for quarks (antiquarks). The longitudinal structure function $F_L = F_2 - 2xF_1$ vanishes at LO (Callan–Gross relation) but receives $\mathcal{O}(\alpha_s)$ NLO corrections.

9.4 The POWHEG method

POWHEG generates the first (hardest) QCD emission with NLO accuracy and provides it as input to a subsequent parton shower. The starting point is the NLO-corrected Born weight

$$\bar{B}(\Phi_n) = B(\Phi_n) + V(\Phi_n) + \int \mathrm{d}\Phi_{\mathrm{rad}}\, C(\Phi_{n+1}), \tag{9.13}$$

where Φ_n is the n-body Born phase space, $B(\Phi_n)$ is the Born squared amplitude, $V(\Phi_n)$ is the renormalized one-loop virtual correction, and $C(\Phi_{n+1})$ is the integrated infrared counterterm that removes double-counting with the real emission. The POWHEG master formula for the fully-differential cross section is

$$\mathrm{d}\sigma_{\mathrm{POWHEG}} = \bar{B}(\Phi_n)\,\mathrm{d}\Phi_n \left[\Delta(\Phi_n,\, p_{T,\mathrm{min}}) + \sum_\alpha \frac{R_\alpha(\Phi_{n+1})}{B(\Phi_n)}\,\Delta(\Phi_n,\, k_{T,\alpha})\,\mathrm{d}\Phi_{\mathrm{rad},\alpha}\right], \tag{9.14}$$

where the sum is over all QCD singular regions α, R_α is the real squared amplitude in region α, and $k_{T,\alpha}$ is the emission transverse momentum in that region. The POWHEG Sudakov form factor is

$$\Delta(\Phi_n,\, p_T) = \exp\left[-\sum_\alpha \int_{p_T}^{\infty} \frac{R_\alpha(\Phi_{n+1})}{B(\Phi_n)}\,\mathrm{d}\Phi_{\mathrm{rad},\alpha}\right]. \tag{9.15}$$

The Sudakov factor ensures that the hardest emission is generated with the correct NLO-improved distribution. Events generated by equation (9.14) carry predominantly positive weights—the property that gives POWHEG its name—and reproduce the NLO-accurate inclusive rate after integration.

For the `nudis` process the Born phase space Φ_n covers the two-body $\nu_\ell q_i \to \ell^- q_j$ kinematics parameterized by (x, y, Q^2). The NLO real corrections include quark-gluon emission

$$\nu_\ell + q_i \longrightarrow \ell^- + q_j + g \tag{9.16}$$

and the gluon-splitting channel

$$\nu_\ell + g \longrightarrow \ell^- + q_j + \bar{q}_i, \tag{9.17}$$

both of which can produce a gluon or an extra $q\bar{q}$ pair among the outgoing hard-process particles in the LHE record.

After the hardest emission is fixed by the POWHEG Sudakov, the parton shower (Pythia 8, not yet invoked in the current HADROS3 pipeline) adds further softer radiation using the POWHEG emission scale as its starting scale.

9.5 Center-of-mass energy and scale

Each accepted DIS interaction vertex in HADROS3 provides the local-frame neutrino energy $E_{\nu,\mathrm{local}}$ to POWHEG as `ebeam1`. The invariant mass of the neutrino–nucleon hard system is

$$\sqrt{s} = \sqrt{2\, m_N\, E_{\nu,\mathrm{local}}}. \tag{9.18}$$

For the UHE range $E_{\nu,\mathrm{local}} \sim 10^9$–$10^{12}$ GeV, this gives $\sqrt{s} \sim 43$–1400 TeV. The upper kinematic bound on the virtuality integration is

$$Q_{\max} = \min\Big(2\sqrt{m_N\, E_{\nu,\mathrm{local}}},\ 10^5\,\mathrm{GeV}\Big)\,, \tag{9.19}$$

and the lower bound is fixed at $Q_{\min} = 10\,\mathrm{GeV}$. The cap at 10^5 GeV prevents integration beyond kinematic regions where the fixed-order NLO approximation is reliable. Renormalization and factorization scales are set equal to the virtuality Q at each phase-space point (`runningscales`$=1$):

$$\mu_R = \mu_F = Q. \tag{9.20}$$

9.6 Parton distributions and CKM flavor structure

HADROS3 uses NNPDF31_nlo_as_0118 (LHAPDF set 303400) for both incoming beams. This is an NLO PDF set consistent with the perturbative order of the POWHEG hard process. The strong coupling α_s is taken from the same PDF set at each renormalization scale (`alphas_from_pdf`$=1$).

For CC $\nu_\ell N$ DIS the incoming parton must be a down-type quark. The weak transition to an up-type quark is weighted by the squared CKM elements. The dominant channels and their approximate weights are

$$d \to u: \quad |V_{ud}|^2 \approx 0.947, \tag{9.21}$$

$$s \to c: \quad |V_{cs}|^2 \approx 0.947, \tag{9.22}$$

$$b \to t: \quad |V_{tb}|^2 \approx 1.0 \quad \text{(kinematically suppressed by } m_t \approx 173\,\mathrm{GeV}\text{)}. \tag{9.23}$$

Off-diagonal CKM contributions ($d \to c$, $s \to u$, etc.) are suppressed by the Cabibbo angle and are included by POWHEG but are numerically subdominant. Antiquark sea contributions, including $\bar{u} \to \bar{d}$ and $\bar{c} \to \bar{s}$ transitions, also enter through the PDF at small x. At NLO, the gluon PDF contributes through the splitting subprocess $\nu_\ell + g \to \ell^- + q_j + \bar{q}_i$. The resulting outgoing parton flavors that can appear in the LHE hard-process record are u, c, d, s, and g, with the exact flavor mix determined by PDF support and POWHEG matrix-element channel selection at the given (x, Q^2).

9.7 LHE hard-process event structure

The Les Houches Event (LHE) file written by `pwhg_main` contains one `<event>` block per generated hard-process realization. Each block records the incoming and outgoing particles with their PDG identifiers, four-momenta (p_x, p_y, p_z, E, m) in GeV, and a status code: -1 for incoming, $+1$ for outgoing. The event header also carries the POWHEG event weight and the hardest-emission scale μ_{POWHEG} that will serve as the starting scale for the subsequent parton shower.

At Born level the event topology is

$$\begin{aligned} \text{status} = -1 : &\quad \nu_\ell + q_i, \\ \text{status} = +1 : &\quad \ell^- + q_j. \end{aligned} \tag{9.24}$$

With the NLO real emission included, the topologies are

$$\begin{aligned} \text{status} = -1 : &\quad \nu_\ell + q_i, \\ \text{status} = +1 : &\quad \ell^- + q_j + g \end{aligned} \qquad \text{or} \qquad \begin{aligned} \text{status} = -1 : &\quad \nu_\ell + g, \\ \text{status} = +1 : &\quad \ell^- + q_j + \bar{q}_i. \end{aligned} \tag{9.25}$$

The event weight in the LHE header is the POWHEG weight from equation (9.14) in units of pb. The total event rate is recovered by summing weights over all events. Because POWHEG weights are predominantly positive, unweighted event generation is possible with statistical efficiency close to unity.

The LHE particles are parton-level hard-process records. PYTHIA 8 hadronization and GEANT4 detector transport are not invoked in the current HADROS3 pipeline (H3-W9b). These records represent the input to future shower and detector stages, not observable hadronic final states.

9.8 Dry-run role

H3-W9a prepares local POWHEG DIS jobs but deliberately does not execute `pwhg_main`. It consumes primary Observer Image Branch records and writes its own products under `POWHEG/`. It does not modify `ObserverBridge/`, `ObserverImageBranches/`, `DIS/`, `ForwardGeodesics/`, or `UHEsource/`.

The H3-W9a run mode is

$$\texttt{powheg\_run\_mode} = \texttt{dry\_run}. \tag{9.26}$$

The reports record

$$\texttt{powheg\_invoked} = \texttt{false}, \tag{9.27}$$

$$\texttt{pwhg\_main\_executed} = \texttt{false}, \tag{9.28}$$

$$\texttt{powheg\_lhe\_generated} = \texttt{false}. \tag{9.29}$$

H3-W9b adds a controlled real-smoke mode:

$$\texttt{powheg\_run\_mode} = \texttt{real\_smoke}. \tag{9.30}$$

This mode executes only the first primary branch record, writes one input card, executes the self-contained local executable `external/powheg/build/DIS/pwhg_main`, and validates a minimal LHE file. It is a pipeline smoke execution, not the full production event-generation stage. PYTHIA, GEANT4, photon transport, and spectra remain disabled.

9.9 POWHEG input branch contract

The driver reads `ObserverImageBranches/observer_image_primary_branches.jsonl`. The selection of downstream candidates is performed by the Observer Bridge and the selection of the primary observed image is performed by H3-W8b. POWHEG does not rank candidates and does not choose among multiple image branches. It prepares one POWHEG request for each primary branch record, except in real-smoke mode where the Python wrapper intentionally executes only the first prepared request as a safety smoke test.

9.10 Input-card mapping

Each selected candidate receives a request identifier of the form `H3PWHG-000001`. Each request has its own `powheg_input_cards/<request_id>/powheg.input`. The implemented mapping is

$$
\begin{aligned}
E_{\nu,\text{local}} &\to \texttt{ebeam1}, &(9.31)\\
m_N = 0.938272\ \text{GeV} &\to \texttt{ebeam2}, &(9.32)\\
\texttt{events\_per\_candidate} &\to \texttt{numevts}, &(9.33)\\
\texttt{channel\_type} &= 3, &(9.34)\\
\texttt{vtype} &= 2. &(9.35)
\end{aligned}
$$

The dry-run card also records fixed PDF choices `lhans1 = 303400` and `lhans2 = 303400`. The implemented scale cap is

$$
Q_{\max} = \min\left(2\sqrt{0.938272\, E_{\nu,\text{local}}},\ 10^5\right). \tag{9.36}
$$

For the implemented seed mode, the seed is

$$
\texttt{powheg\_seed} = \texttt{random\_seed} + \texttt{candidate\_rank}. \tag{9.37}
$$

9.11 Real-smoke and real-free LHE validation

In H3-W9b real-smoke mode, the prepared POWHEG card is copied into an isolated work directory and `pwhg_main` is executed with that directory as its working directory. The produced `pwgevents.lhe` must satisfy

$$
\begin{aligned}
\texttt{<LesHouchesEvents} &\in \texttt{pwgevents.lhe}, &(9.38)\\
\texttt{</LesHouchesEvents>} &\in \texttt{pwgevents.lhe}, &(9.39)\\
N_{\text{LHE events}} &> 0. &(9.40)
\end{aligned}
$$

The validated LHE and run log are copied to

$$
\begin{aligned}
&\texttt{POWHEG/powheg\_lhe/H3PWHG-000001/pwgevents.lhe}, &(9.41)\\
&\texttt{POWHEG/powheg\_run\_logs/H3PWHG-000001/powheg.log}. &(9.42)
\end{aligned}
$$

The validation report records `powheg_invoked=true`, `pwhg_main_executed=true`, `powheg_lhe_generated=tru` and `n_lhe_events`. This smoke mode does not modify `ObserverImageBranches/`, `ObserverBridge/`, `DIS/`, `ForwardGeodesics/`, or `UHEsource/`.

H3-W9b real-free mode uses the same local POWHEG execution and LHE validation rules, but it removes the real-smoke safety clamp. The number of interaction candidates is set by the number of records in `ObserverImageBranches/observer_image_primary_branches.jsonl`, and the number of independent POWHEG hard-scattering realizations per selected interaction is set by `events_per_candidate`. Each primary branch record is executed in its own isolated POWHEG work directory. The produced LHE files are copied under `POWHEG/powheg_lhe/<powheg_request_id>/pwgevent` and the parser aggregates all produced LHE files into the POWHEG particle and event summaries. The provenance records `powheg_real_free_invoked=true`, `real_free_mode=true`, `n_powheg_jobs_requested`, `n_powheg_jobs_run`, `events_per_candidate_requested`, and `n_lhe_events_tota` This mode still does not invoke PYTHIA, GEANT4, photon transport, or spectra.

9.12 Event requests

The event request file `POWHEG/powheg_event_requests.jsonl` records for each prepared job:

$$\begin{aligned}\mathcal{R}_{\mathrm{POWHEG}} = \{&\texttt{powheg\_request\_id},\ \texttt{candidate\_rank},\ \texttt{interaction\_id},\ \texttt{event\_id},\ \texttt{source\_sample\_id},\\ &E_{\nu,\mathrm{local}},\ w_{\mathrm{physics}},\ w_{\mathrm{observer}},\ S_{\mathrm{obs}},\ \texttt{powheg\_input\_path},\ \texttt{powheg\_seed},\ \texttt{powheg\_status}\}.\end{aligned} \tag{9.43}$$

The dry-run status is `dry_run_ready`; `powheg_invoked` is false for every dry-run request. In real-smoke mode the initial status is `real_smoke_ready`; after successful validation the request records `real_smoke_lhe_generated`, `powheg_invoked=true`, `pwhg_main_executed=true`, the LHE path, the log path, and `n_lhe_events`. In real-free mode the analogous statuses are `real_free_ready` and `real_free_lhe_generated`; multiple validated LHE products may be recorded in a single POWHEG run.

9.13 Inputs and outputs

Role	Implemented item
Inputs	`ObserverImageBranches/observer_image_primary_branches.jsonl` and POWHEG configuration values.
Outputs	`POWHEG/powheg_event_requests.jsonl`, input cards, summary JSON/CSV, report JSON, validation report, diagnostic plots, and in real-smoke or real-free mode validated `pwgevents.lhe` products.
Still disabled	No PYTHIA; no GEANT4; no photon transport; no spectra.
Backend	C++17 executable `bin/hadros3_powheg_driver`. Python handles orchestration and plots.

Chapter 10

Provenance and Statistical Weights

10.1 Stage provenance

Each stage records the backend, products, validation flags, and disabled future stages relevant to its current implementation. A typical provenance record contains:

$$\{\text{parameters}, \text{products}, \text{validation}, \text{stage summaries}, \text{disabled stages}\}. \tag{10.1}$$

The implemented provenance explicitly marks future expensive stages as disabled until they are introduced:

$$\texttt{pythia\_invoked} = \texttt{false}, \tag{10.2}$$
$$\texttt{geant4\_invoked} = \texttt{false}, \tag{10.3}$$
$$\texttt{photon\_transport\_invoked} = \texttt{false}, \tag{10.4}$$
$$\texttt{expensive\_event\_generation\_invoked} = \texttt{false}. \tag{10.5}$$

For H3-W9a, POWHEG is active only as dry-run preparation:

$$\texttt{status} = \texttt{powheg\_jobs\_prepared\_no\_lhe}. \tag{10.6}$$

For H3-W9b real-smoke execution:

$$\texttt{status} = \texttt{powheg\_real\_smoke\_lhe\_generated}. \tag{10.7}$$

The provenance also records the scientific theory reference for the run:

$$\texttt{theory\_document} = \texttt{docs/Theory/HADROS3\_Physics\_Theory.pdf}, \tag{10.8}$$
$$\texttt{theory\_version} = \texttt{\textbackslash TheoryVersion }, \tag{10.9}$$
$$\texttt{theory\_commit} = \texttt{runtime git commit}, \tag{10.10}$$
$$\texttt{theory\_generation\_date} = \texttt{runtime provenance timestamp}. \tag{10.11}$$

These fields connect each generated result to the theory document that describes the implemented physical and statistical assumptions.

10.2 Weight chain

The implemented source and direction samplers are configured so that the default source and direction weights are unity:

$$w_{\text{source}} = 1, \tag{10.12}$$
$$w_{\text{direction}} = 1. \tag{10.13}$$

The DIS sampler records interaction and pre-event weights. The Observer Bridge then uses

$$w_{\text{physics}} = \texttt{final\_pre\_event\_weight} \tag{10.14}$$

and computes

$$S_{\text{obs}} = w_{\text{physics}} w_{\text{observer}}. \tag{10.15}$$

The POWHEG dry-run and real-smoke modes carry those weights into each event request. They do not change the physics score. H3-W9a stops after card preparation; H3-W9b executes one local POWHEG smoke job and validates the resulting minimal LHE.

10.3 Random seeds

The implemented random stages record seeds:

- UHE source sampling records its source seed.
- DIS interaction sampling records its acceptance and point-sampling seed.
- POWHEG dry-run and real-smoke modes record `random_seed` and the per-candidate `powheg_seed`.

These seeds are part of the statistical contract of the pipeline.

Chapter 11

Pipeline Summary

11.1 Implemented stage contract

The current implemented pipeline can be summarized as

$$\text{UHESource} \to \text{ForwardGeodesics} \to \text{DIS} \to \text{ObserverBridge} \to \text{POWHEG RealSmoke}. \tag{11.1}$$

The camera preview is an observer-view diagnostic that can be used by the dashboard and by Observer Bridge overlays, but it is not a physical secondary-particle propagation stage.

Table 11.1: Implemented pipeline products and semantic status.

Stage	Official inputs	Official outputs	Status
UHE Source	Configuration	`UHEsource/` JSONL/JSON	Sampling proxy with monoenergetic source.
Forward Geodesics	`UHEsource/`	`ForwardGeodesics/` paths and segments	Kerr null geodesic propagation.
DIS Sampler	`ForwardGeodesics/`, `UHEsource/`	`DIS/` optical-depth and accepted interactions	Optical-depth DIS sampler.
Observer Bridge	`DIS/dis_accepted_interactions.jsonl`	`ObserverBridge/` candidates and ranked events	Non-destructive geometric scoring proxy.
Observer Image Branches	`ObserverBridge/` selected candidates and Kerr pixel-map products	`ObserverImageBranches/` branches and primary branches	Observed image branch proxy selection.
POWHEG Real Smoke	`ObserverImageBranches/observer_image_primary_branches.jsonl`	`POWHEG/` requests, run-input cards, validation report, and one smoke LHE	H3-W9a dry run plus H3-W9b local one-job LHE smoke.

11.2 Explicit non-goals of the implemented stages

The current pipeline does not yet implement:

- production-scale POWHEG event execution beyond the real-smoke mode;
- PYTHIA showering and hadronization;
- GEANT4 detector or material transport;
- photon transport from secondaries to the observer;
- full GRMHD torus evolution;
- full physical visibility, escape, redshift, or line-of-sight transport.

Those future stages must preserve the contract that each stage consumes official previous outputs, writes only to its own directory, records provenance, and marks expensive or proxy physics explicitly.

Appendix A

Appendix: Symbols and Product Names

Product or field	Meaning
uhe_neutrino_source_samples.jsonl	UHE source positions, energies, directions, and sampling weights.
uhe_neutrino_forward_path_segments.jsonl	Forward geodesic path segments with midpoint coordinates, momenta, and segment lengths.
dis_path_optical_depths.jsonl	Path-level DIS optical depths and interaction probabilities.
dis_accepted_interactions.jsonl	Accepted DIS interactions and their density-checked interaction points.
observer_bridge_candidates.jsonl	All accepted DIS interactions scored as observation candidates.
observer_bridge_ranked_events.jsonl	Same candidate family ranked by final observation score.
observer_image_primary_branches.jsonl	Primary observed image branch per selected Observer Bridge candidate.
powheg_event_requests.jsonl	Dry-run POWHEG event requests and card paths.
final_pre_event_weight	Physics weight consumed by Observer Bridge as implemented.
final_observation_score	Product of implemented physics and observer weights.
powheg_invoked	False in H3-W9a dry run; true in H3-W9b real smoke.
pwhg_main_executed	False in H3-W9a dry run; true in H3-W9b real smoke.
powheg_lhe_generated	False in H3-W9a dry run; true in H3-W9b real smoke.